

### Guidelines:

You should work on these exercises during the first two and a half weeks of the course. At the end of the third week, you will take a "Homework Oriented Exam" (see Canvas/Datanose for details). The exam will include exercises similar to those in this homework set, possibly with minor variations or additional questions.

We will not provide a solution sheet for this homework set. However, you are encouraged to discuss any questions you have about the exercises with your TAs during the Friday seminars.

When working on your answers, please remember the following points:

- Your solutions should include the intermediate steps of your reasoning and derivations. Answers without justification will not receive full marks in the exam. Also, remember to simplify your final answers as much as possible.
- We use the numerator layout convention for multivariate derivatives (i.e., the derivative of a function  $f : \mathbb{R}^m \rightarrow \mathbb{R}^n$  has shape  $n \times m$ ).
- When computing multivariate derivatives, you should first compute the derivative for arbitrary components (e.g.,  $\frac{\partial f_i}{\partial x_j}$ ). After deriving this general form, assemble the final answer as a matrix/vector and simplify. This is what we refer to as *index notation*.

### Hand-in Exercises

You may submit the following exercises on the dates below to receive feedback from your TAs:

- Friday, September 4th: Exercise 1e
- Friday, September 11th: Exercise 3

Please note:

- Only handwritten, paper submissions will be accepted.
- Please write your name and group (1,2,3/4,5,6/...) in the top right corner.
- Submission is voluntary and will not affect your grade. Please ensure your submitted solutions are entirely your own work, as this allows us to provide the most helpful feedback.

### Exercise 1: Multivariate Calculus

In this exercise, you are going to compute several gradients. Simplify your answers as much as possible, and use *index-notation* for your derivations where appropriate.

- $\frac{d}{dx} \tanh(\mathbf{x})$  with  $\mathbf{x} \in \mathbb{R}^m$ , where  $\tanh$  is the hyperbolic tangent function applied element-wise.
- $\frac{d}{d\mathbf{w}} f$  with  $f = (\mathbf{X}\mathbf{w})^T(\mathbf{X}\mathbf{w})$ , where  $\mathbf{X} \in \mathbb{R}^{m \times n}$  and  $\mathbf{w} \in \mathbb{R}^n$ .
- $\frac{d}{d\mathbf{w}} f$  with  $f = \mathbf{w}^T \mathbf{X} \mathbf{w}$ , where  $\mathbf{X} \in \mathbb{R}^{n \times n}$  and  $\mathbf{w} \in \mathbb{R}^n$ .
- $\frac{d}{dx} \varsigma(\mathbf{x})$  with  $\mathbf{x} \in \mathbb{R}^n$ , where  $\varsigma$  is the softmax function  $\varsigma(\mathbf{x})_i = \frac{\exp(x_i)}{\sum_{k=1}^n \exp(x_k)}$ .
- $\frac{d}{d\theta} \frac{1}{n} \|\mathbf{X}\theta - \mathbf{y}\|_2^2$ , with  $\mathbf{X} \in \mathbb{R}^{n \times d}$ ,  $\theta \in \mathbb{R}^d$ ,  $\mathbf{y} \in \mathbb{R}^n$ . Set this to zero and solve for  $\theta$  to compute the stationary point of the function, and prove that it is indeed a minimum.

## Exercise 2: Full analysis of a distribution: Poisson distribution

The Poisson process is a model for a series of discrete events where the average number of events in a fixed interval is known. It is also assumed that the process is memoryless or Markovian, i.e. the occurrence of a new event is independent of the previous events. In this exercise, we are interested in analyzing the Poisson distribution, which is a discrete probability distribution that expresses the probability of a given number of events occurring in a fixed interval of time or space. There are many processes that are modeled this way. For example, it is used to model the number of radioactive decays in a fixed time period, the number of calls arriving at a call center per hour, or the number of typos on a page. Formally, a Poisson distribution with rate parameter  $\lambda > 0$  can be modeled as such:

$$p(k; \lambda) = \frac{\lambda^k e^{-\lambda}}{k!}, \quad k \in \{0, 1, 2, \dots\}$$

The aim of this exercise is to familiarize you with arbitrary distributions. Note that by no means the following questions are the only things you might want to know about a distribution, but rather serve as a starting point for further research. Using these insights, answer the following questions:

- (a) If  $K \sim \text{Poisson}(\lambda)$ , prove that  $\mathbb{E}[K] = \lambda$ .
- (b) A customer support center receives an average of 4 calls per hour. Assuming the number of calls follows a Poisson distribution, what is the probability of receiving exactly 3 calls in a given hour?
- (c) Consider a dataset  $\mathbf{k} = [k_1, k_2, \dots, k_N]$  which are independent and identically distributed (i.i.d.) random variables from a Poisson distribution with the rate parameter  $\lambda$ . Derive the log-likelihood function for the parameter  $\lambda$  given the dataset.
- (d) Find the maximum likelihood estimator  $\lambda_{\text{ML}}$  for the likelihood function calculated in part c).
- (e) Assume that the prior for the parameter  $\lambda$  is given by the Gamma distribution with hyperparameters  $\alpha_1$  and  $\alpha_2$ :

$$p(\lambda | \alpha_1, \alpha_2) = \frac{\alpha_2^{\alpha_1}}{\Gamma(\alpha_1)} \lambda^{\alpha_1-1} e^{-\alpha_2 \lambda},$$

where  $\Gamma$  denotes the gamma function. Show that we can find  $\lambda_{\text{MAP}}$  by optimizing:

$$\lambda_{\text{MAP}} = \arg \max_{\lambda} \left( \sum_{i=1}^N k_i + \alpha_1 - 1 \right) \log \lambda - (N + \alpha_2) \lambda.$$

**Hint:** Show first that  $\lambda_{\text{MAP}} = \arg \max_{\lambda} \log p(\mathbf{k} | \lambda) + \log p(\lambda)$ .

- (f) Find the MAP estimator  $\lambda_{\text{MAP}}$ .
- (g) In the case of a Poisson distribution with a Gamma prior, the resulting posterior distribution can be derived analytically. Show that the posterior distribution is indeed a Gamma distribution.

**Hint:** The resulting distribution follows  $\text{Gamma}(\alpha'_1, \alpha'_2)$  with  $\alpha'_1 = \sum_{i=1}^N k_i + \alpha_1$  and  $\alpha'_2 = N + \alpha_2$ .

## Exercise 3: Counting Fish

Imagine you are in an aquarium, and you notice that each fish has a tag with a number attached to its fin. You see the numbers 8, 2, 14, 5, and then 14 again. After asking an employee, they tell you that each fish has a consecutive number assigned, and that the sequence starts at one.

- (a) Intuitively, how many fishes do you think there are after observing these five samples?
- (b) What is the most likely estimate? State your assumptions clearly.

**Hint:** To find the "most likely" estimate, you first need a likelihood function. This requires assuming a probability distribution for the observed tags. If you have no reason to believe one fish tag is more likely to be seen than another, which distribution should you use?

- (c) The MLE estimator you derived in the previous step is the maximum of the observed samples. Let's denote this estimator by  $\hat{M} = X_{\max}$ . It is known to underestimate the true value. An adjusted estimator that attempts to correct this is  $\hat{M}_{adj} = \frac{n+1}{n} X_{\max} - 1$ . Compute the bias of this adjusted estimator.

**Hint:** The bias is  $\mathbb{E}[\hat{M}_{adj}] - M$ . Use the linearity of expectation to find  $\mathbb{E}[\hat{M}_{adj}]$ . You may use the following known result for the expected value of the maximum of  $n$  samples from a discrete uniform distribution  $\{1, \dots, M\}$  without proof:

$$\mathbb{E}[X_{\max}] = M - \frac{1}{M^n} \sum_{x=0}^{M-1} x^n$$

Fun fact: The Allies used a similar statistical technique during WWII to estimate the monthly production rate of German tanks based on the serial numbers of captured or destroyed tanks. This estimation was significantly more accurate than intelligence reports.

#### Exercise 4: General Multiple Outputs Linear Regression

So far, our linear regression models have assumed that the target  $t$  is a single scalar value. In a more general case, we may wish to predict a  $K$ -dimensional target vector  $\mathbf{t}$ . A common approach is to use the same set of basis functions to model all components of the target vector, leading to a model of the form:

$$\mathbf{y}(\mathbf{x}, \mathbf{W})^\top = \phi(\mathbf{x})^\top \mathbf{W},$$

where  $\mathbf{y}$  is the  $K$ -dimensional model prediction,  $\mathbf{x}$  is the input vector,  $\phi(\mathbf{x})$  is an  $M$ -dimensional feature vector (with  $\phi_0(\mathbf{x}) = 1$  by convention), and  $\mathbf{W}$  is the parameter matrix. We assume the conditional distribution of the target vector is a spherical Gaussian:

$$p(\mathbf{t} | \mathbf{x}, \mathbf{W}, \sigma^2) = \mathcal{N}(\mathbf{t} | \mathbf{y}(\mathbf{x}, \mathbf{W}), \sigma^2 \mathbf{I}),$$

where  $\sigma^2$  is the variance and  $\mathbf{I}$  is the  $K \times K$  identity matrix. Given a dataset of  $N$  i.i.d. observations  $\{(\mathbf{x}_n, \mathbf{t}_n)\}_{n=1}^N$ , we can group the targets into an  $N \times K$  matrix  $\mathbf{T}$  and the feature vectors into an  $N \times M$  design matrix  $\Phi$ .

- (a) What are the dimensions of the parameter matrix  $\mathbf{W}$ ?
- (b) Write down the log-likelihood function for the parameters  $\mathbf{W}$  and  $\sigma^2$ , given the dataset  $(\Phi, \mathbf{T})$ .
- (c) Find the maximum likelihood solution  $\mathbf{W}_{\text{ML}}$  and show that it is independent of the variance  $\sigma^2$ .
- (d) Show that the maximum likelihood solution for  $\sigma^2$  is given by:

$$\sigma_{\text{ML}}^2 = \frac{1}{NK} \|\mathbf{T} - \Phi \mathbf{W}_{\text{ML}}\|_F^2$$

### Exercise 5: Bayesian Linear Regression

Consider a linear basis function model (such as in Bishop 3.1), and suppose that we have already observed  $N$  data points so that the posterior distribution over  $\mathbf{w}$  for the observed data ( $N$ ) is given by:

$$p(\mathbf{w}|\mathbf{t}) = \mathcal{N}(\mathbf{w}|\boldsymbol{\mu}_N, \boldsymbol{\Sigma}_N) ,$$

where

$$\boldsymbol{\mu}_N = \boldsymbol{\Sigma}_N(\boldsymbol{\Sigma}_0^{-1}\boldsymbol{\mu}_0 + \beta\boldsymbol{\Phi}_N^T\mathbf{t}_N) ,$$

and

$$\boldsymbol{\Sigma}_N^{-1} = \boldsymbol{\Sigma}_0^{-1} + \beta\boldsymbol{\Phi}_N^T\boldsymbol{\Phi}_N .$$

When a new data-point,  $(\mathbf{x}_{N+1}, t_{N+1})$ , is observed, this posterior can be regarded as the prior for the next observation. By considering this additional data-point, show that the resulting posterior distribution is equal to:

$$p(\mathbf{w}|\mathbf{t}) = \mathcal{N}(\mathbf{w}|\boldsymbol{\mu}_{N+1}, \boldsymbol{\Sigma}_{N+1}) ,$$

with

$$\boldsymbol{\mu}_{N+1} = \boldsymbol{\Sigma}_{N+1}(\boldsymbol{\Sigma}_N^{-1}\boldsymbol{\mu}_N + \beta\boldsymbol{\phi}_{N+1}t_{N+1}) ,$$

and

$$\boldsymbol{\Sigma}_{N+1}^{-1} = \boldsymbol{\Sigma}_N^{-1} + \beta\boldsymbol{\phi}_{N+1}\boldsymbol{\phi}_{N+1}^T .$$

To do this, answer the following questions:

- Which prior distribution  $p(\mathbf{w}|\mathbf{S}_0, \mathbf{m}_0)$  has been used to obtain the reported posterior? Write your answer as a function of  $\boldsymbol{\Sigma}_0$  and  $\boldsymbol{\mu}_0$ .  
 Hint: What happens to the posterior distribution when no data is observed?
- What happens to the posterior update rule when the precision  $\beta$  approaches 0? Explain with your own words.
- When a new example,  $(\boldsymbol{\phi}_{N+1}, t_{N+1})$ , is observed, the posterior  $p(\mathbf{w}|\boldsymbol{\Phi}_N, \mathbf{t}_N, \boldsymbol{\Sigma}_0, \boldsymbol{\mu}_0, \beta)$  can be regarded as the prior for the next observation. Write down the expression for the updated posterior after observing the new data-point  $p(\mathbf{w}|\boldsymbol{\Phi}_{N+1}, \mathbf{t}_{N+1}, \boldsymbol{\Sigma}_0, \boldsymbol{\mu}_0, \beta)$  as a function of the old posterior, likelihood of the new observation and its evidence.
- By expanding the expression reported in the previous answer, show that the updated posterior distribution is equal to:

$$p(\mathbf{w}|\boldsymbol{\Phi}_{N+1}, \mathbf{t}_{N+1}, \boldsymbol{\Sigma}_0, \boldsymbol{\mu}_0, \beta) = \mathcal{N}(\mathbf{w}|\boldsymbol{\mu}_{N+1}, \boldsymbol{\Sigma}_{N+1}) ,$$

with

$$\begin{aligned} \boldsymbol{\mu}_{N+1} &= \boldsymbol{\Sigma}_{N+1}(\boldsymbol{\Sigma}_N^{-1}\boldsymbol{\mu}_N + \beta\boldsymbol{\phi}_{N+1}t_{N+1}) , \\ \boldsymbol{\Sigma}_{N+1}^{-1} &= \boldsymbol{\Sigma}_N^{-1} + \beta\boldsymbol{\phi}_{N+1}\boldsymbol{\phi}_{N+1}^T . \end{aligned}$$

Hint: focus on the terms inside the exponential; you can discard normalization terms which do not depend on  $\mathbf{w}$  during your derivation to identify mean and covariance of the updated posterior.